Charles Shaju

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SUMMARY

Robotics and embedded engineer with 2+ years in marine robotics (ASVs, underwater ROVs), now pursuing an MCA to move into perception and autonomy. Tuned PID control on an 80 kg catamaran ASV used for dam surveys at 4 sites; building a ROS2 mobile robot with EKF sensor fusion on Raspberry Pi and ESP32.

WORK EXPERIENCE

Embedded Systems Engineer 08/2023 – 06/2025
i4 Marine Technologies Pvt Ltd Pune, Maharashtra
Present (part-time since June 2025)

- Tuned steering, speed and acceleration gains on an ArduPilot/Pixhawk catamaran ASV (80 kg payload), using Mission Planner gyro graphs and logs to remove path zigzag; held planned survey lines within 1 m during bathymetry and dam-wall inspection at 4 dams.
- Set up UDP telemetry over radio modems and a live sensor-data UI for the ASV up to 2-3km
- Wired, configured, calibrated and tested underwater ROVs, including a commercial BlueROV.
- Built IoT speed-control modules for electric canoes; currently building GPS IoT modules for [spell out OBM] and a precision barometer.

EdTech & Robotics Solutions Engineer (Intern) 04/2026 – 05/2026
Jobin & Jismi Thrissur, Kerala
Internship

- Built a browser-based block-programming platform (Google Blockly + JavaScript) that connects over Bluetooth to an ESP32 Otto robot, so students can drive it and program it with blocks.
- Tested and debugged the Edu robot's motor control and movements.

Robotics Intern 01/2023 – 07/2023
Shristi Robotics Thrissur, Kerala

- Built a 6-thruster underwater ROV (Pixhawk, ArduSub) on a funded project: wiring, calibration and testing; deployed for inspection
- Operated a 3D printer to produce ROV components.
- Taught mobile robotics, sensors and Arduino: a 2-week summer course for school students and one-day workshops for college students.

PROJECTS

ROS2 differential-drive robot
(Raspberry Pi 4 + ESP32) — custom swept-TFmini 2D lidar (stepper sweep, limit-switch homing), BNO085 IMU and wheel encoders fused with an EKF (robot_localization); teleoperated with live scan and odometry in RViz over WiFi; bring-up and debugging documented in a public repo.

CNC pen plotter
Built an A3-size two-axis Cartesian XY plotter with a servo pen-lift, hand-soldered perfboard electronics and 3D-printed parts; configured FluidNC (GRBL-compatible) on an ESP32; delivered to a client.

Depth-based obstacle avoidance
Python with an Intel RealSense D455, tested driving on a real surface.

SKILLS

Languages
Python, Embedded C/C++ (Arduino, PlatformIO), JavaScript (Blockly), Java, SQL

Sensors & Actuators
Barometric Sensors, Depth, GPS, IMU, Motor Control, pH, Pressure, Turbidity, Temperature, TFmini

Hardware
ESP32, Raspberry Pi, Pixhawk, RealSense D455, Toradex

Robotics
ROS 2 Humble, RViz, ArduPilot/ArduSub, MAVLink, Mission Planner, QGroundControl, PID tuning, EKF sensor fusion, KiCad, OnShape (Basic)

Communication Protocols
I2C, SPI, UART, PWM, LoRa, RS485/Modbus, Wi-Fi, UDP, WiFi, Bluetooth

EDUCATION

MCA in Computer Application 06/2025 – Present
Rajagiri College of Social Science Kochi, Kerala
Course Type: Full-Time

Bachelor of Vocational Studies (IT) 08/2020 – 04/2023
Christ College Autonomous Irinjalakuda Thrissur, Kerala
GPA: 85% Course Type: Full-Time

HSS (Plus Two) in Commerce Computer Application 07/2018 – 04/2020
St. George HSS Pariyaram Thrissur, Kerala
GPA: 97% Course Type: Full-Time

CERTIFICATIONS

MathWorks Computer Vision Engineer
07/2026 MathWorks

Certificate in AI-Machine Learning Developer
06/2024 ASAP Kerala